Customer Shopping Behavior Analysis

Analyzing **3,900 transactions** across 18 features to uncover spending patterns, customer segments, and product preferences — guiding strategic business decisions.

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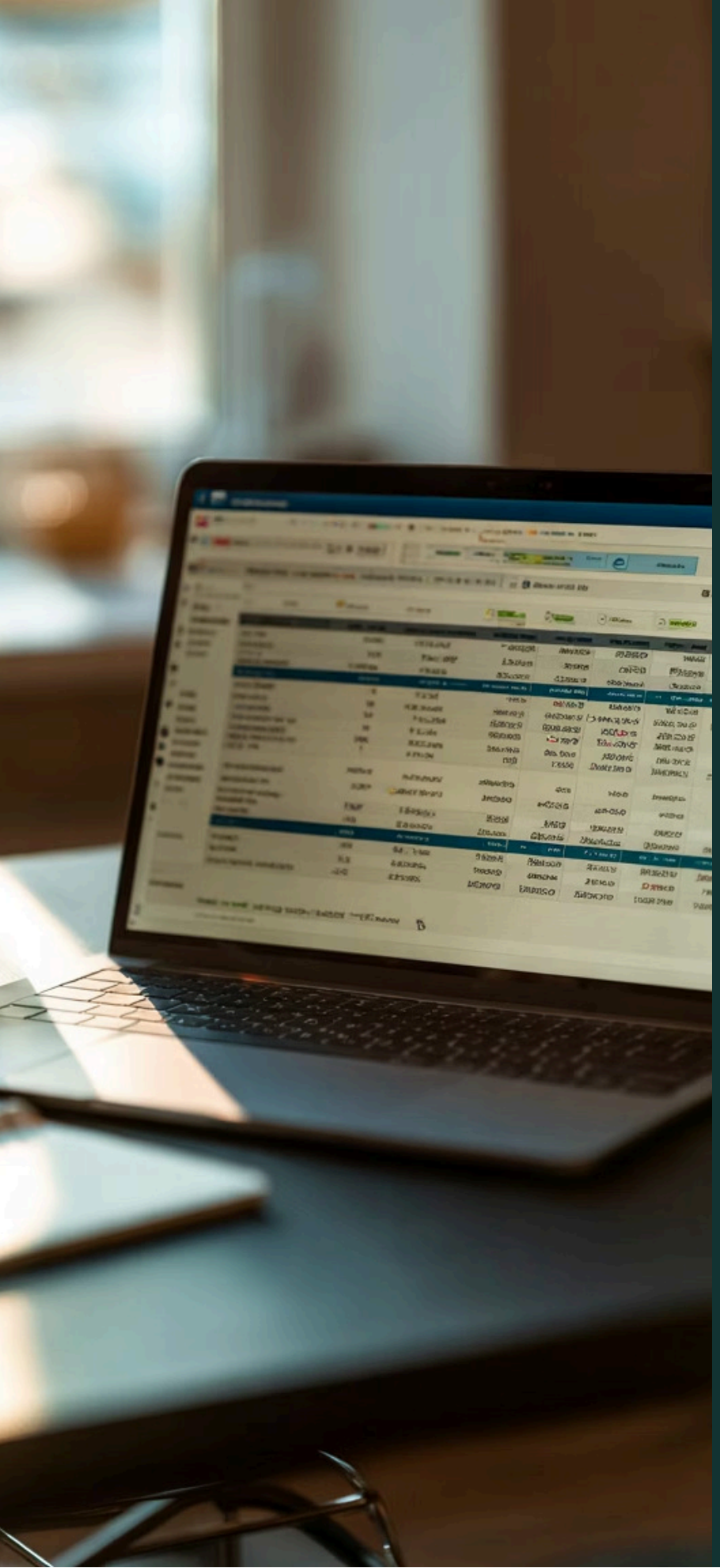

Dataset at a Glance

3,900

Transactions

18

Features

37

Missing Values

Review Rating column only

50

Locations

Features span **customer demographics** (Age, Gender, Location, Subscription Status), **purchase details** (Category, Amount, Season, Size, Color), and **shopping behavior** (Discount, Frequency, Review Rating, Shipping Type).

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Data Preparation in Python

01

Load & Explore

Imported via pandas; used `df.info()` and `df.describe()` for structure and summary stats.

02

Handle Missing Data

Imputed 37 missing Review Ratings using **median per category**.

03

Standardize Columns

Renamed to snake_case; dropped redundant `promo_code_used`.

04

Feature Engineering

Created `age_group` (Young Adult → Senior) and `purchase_frequency_days` from frequency labels.

05

Database Integration

Loaded cleaned DataFrame into **PostgreSQL** via SQLAlchemy for SQL analysis.

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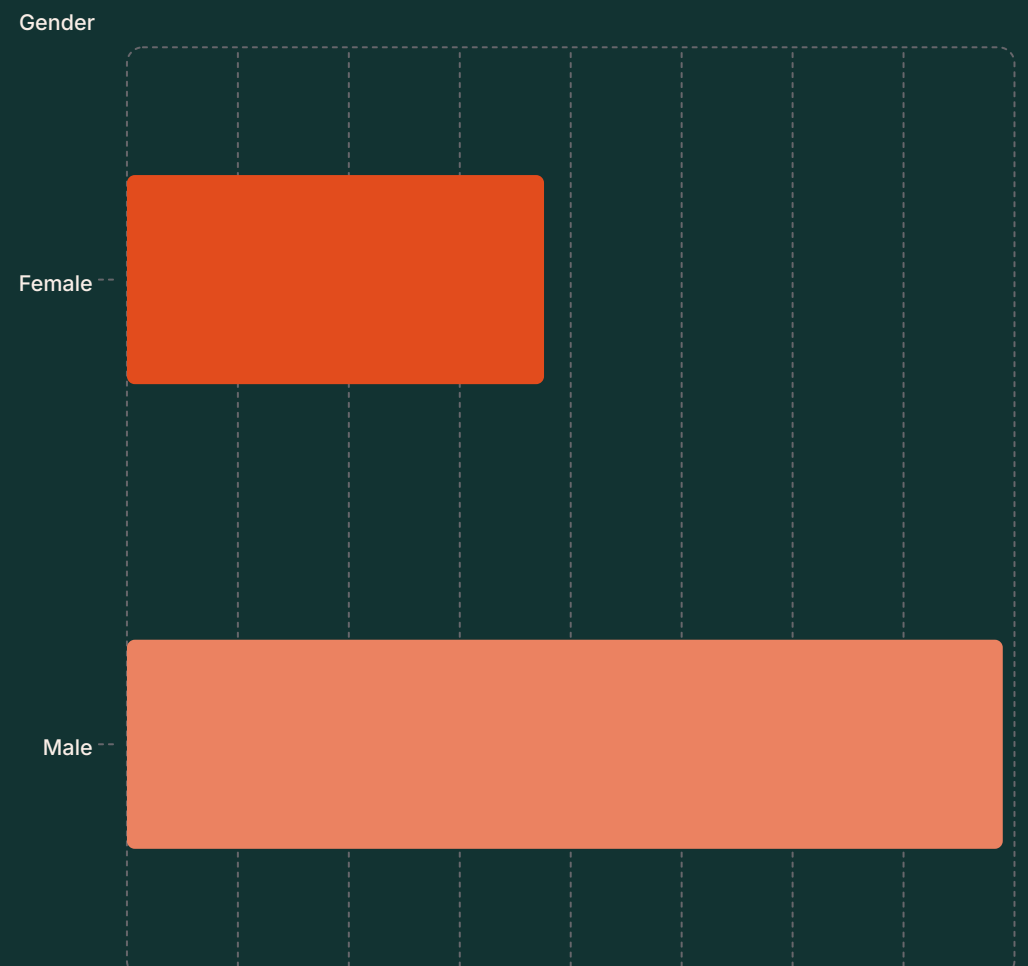
Key Findings: Revenue & Discounts



Revenue by Gender

Male customers generated **$157,890** in revenue — over **2× more** than female customers ($75,191).

Discount-Dependent Products

Top 5 products by discount rate: **Hat (50%)**, Sneakers (49%), Coat (49%), Sweater (48%), Pants (47%).

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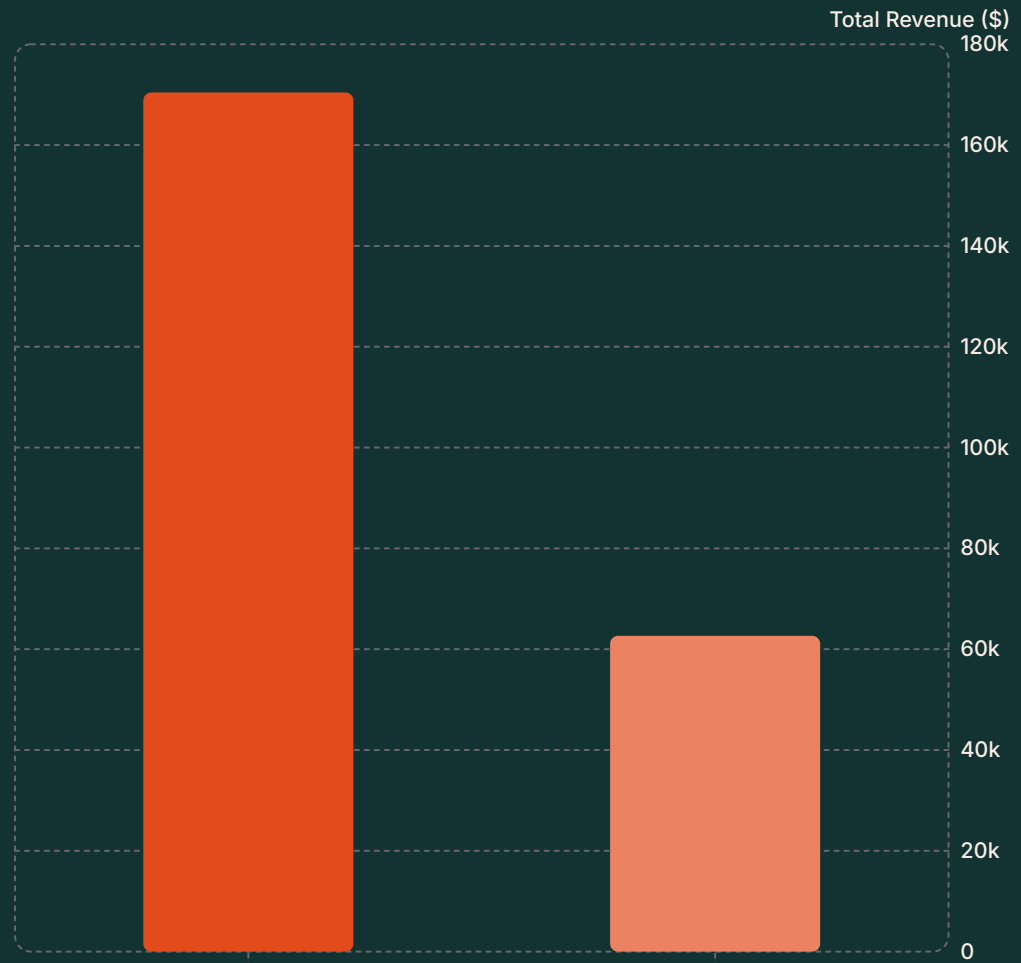
Subscribers vs. Non-Subscribers



Revenue Gap

Non-subscribers drive **$170,436** in revenue vs. subscribers' **$62,645** — despite nearly identical average spend (~$59.50).

Repeat Buyers

Of customers with >**5 previous purchases**, 2,518 are non-subscribers vs. only 958 subscribers — a major conversion opportunity.

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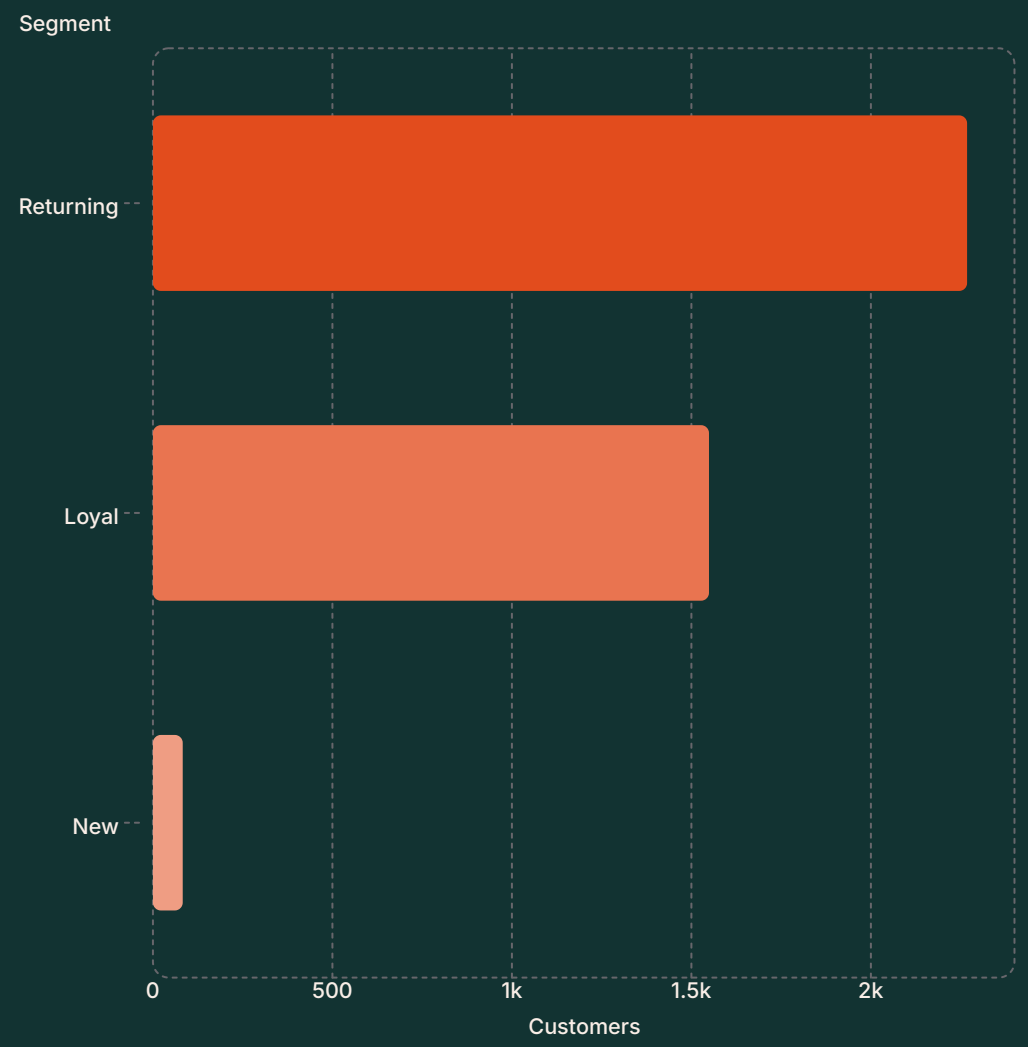
Customer Segmentation



Segment Definitions

New

1 previous purchase — **83 customers**

Returning

2–30 purchases — **2,268 customers**

Loyal

30+ purchases — **1,549 customers**

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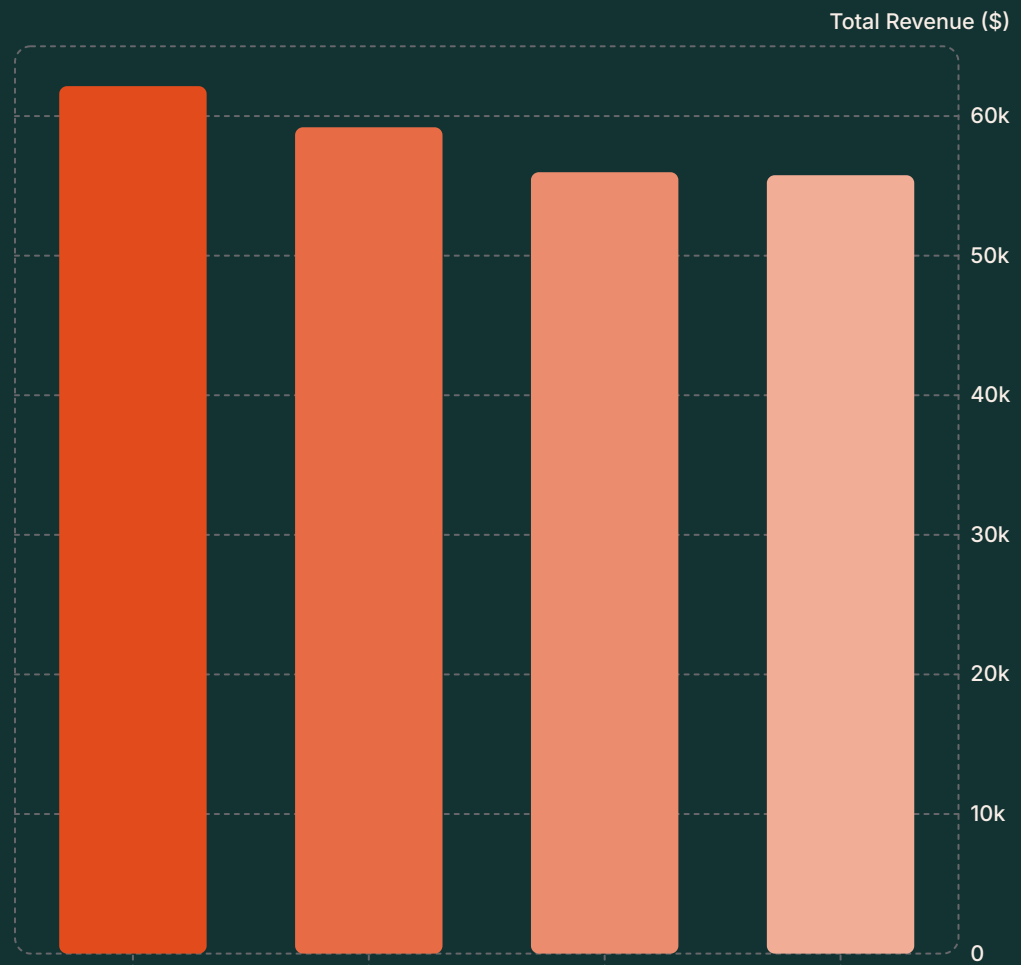
Revenue by Age Group



Revenue Distribution

Young Adults lead with $62,143 in total revenue, followed closely by Middle-Aged ($59,197), Adult ($55,978), and Senior ($55,763) groups.

Revenue is relatively evenly distributed across all age groups — a sign of broad product appeal.

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Top Products by Category

Accessories

Jewelry (171), Sunglasses (161), Belt (161)

Clothing

Blouse (171), Pants (171), Shirt (169)

Footwear

Sandals (160), Shoes (150), Sneakers (145)

Outerwear

Jacket (163), Coat (161)

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Power BI Dashboard

Dashboard Highlights

4K Customers

Avg. purchase $59.76, avg. rating 3.75

Revenue by Category

Clothing 45%, Accessories 32%, Footwear 15%, Outerwear 8%

Subscription Rate

Only 7.3% subscribed — major growth potential

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Business Recommendations

Customer Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment — 1,549 already qualify.

Boost Subscriptions

Promote exclusive subscriber benefits. 2,518 repeat buyers are non-subscribers — a prime conversion target.

Targeted Marketing

Focus on high-revenue age groups (Young Adults, Middle-Aged) and express-shipping users.

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